Insurance Business Intelligence Analytics Report

Data Simulation & Analytics Assignment

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Role: Business Intelligence Internship
Tools Used: Python, Pandas, SQL-style analysis, Power BI
Date: March 2026

INTRODUCTION

This project simulates an insurance portfolio to analyze the relationship between policy premiums and claim costs. Synthetic policy and claims datasets were generated using Python, then cleaned and integrated into an analysis-ready dataset. The data was analyzed and visualized using Power BI to evaluate portfolio risk and profitability.

BUSINESS PROBLEM DESCRIPTION

Business Problem

Insurance companies must monitor their policy portfolios to evaluate risk exposure and financial performance. A key metric used is the loss ratio, which compares total claims to premium revenue.

This project simulates an insurance portfolio to analyze claim behavior, premium revenue, and overall policy profitability under defined business assumptions.

Project Objectives

• Simulate large-scale insurance policy sales data for 2024
• Generate claims data based on predefined claim rules
• Integrate policy and claims datasets into a single dataset
• Clean and prepare the data for analysis
• Perform analytical queries to evaluate claim costs and premium revenue
• Build visualizations to understand claim trends and portfolio profitability

1. Approach and Assumptions

This section describes the methodology used to generate and prepare the datasets for analysis. The process involved dataset simulation, data preparation, and validation.

A. Policy Sales Dataset

A simulated Policy Sales dataset was created to represent vehicle insurance policies purchased in 2024.

Key Characteristics

• Total policies generated: 1,000,000
• Each record represents a unique customer and vehicle
• All vehicles were assigned a value of ₹100,000

Dataset Fields

• Customer_ID – Unique customer identifier
• Vehicle_ID – Unique vehicle identifier
• Vehicle_Value – Vehicle value
• Premium – Policy premium amount
• Policy_Purchase_Date – Policy purchase date
• Policy_Start_Date – Policy activation date
• Policy_End_Date – Policy expiry date
• Policy_Tenure – Policy duration in years

Policy Tenure Distribution

• 20% → 1-year policies
• 30% → 2-year policies
• 40% → 3-year policies
• 10% → 4-year policies

Additional Assumptions

• Policy purchases were evenly distributed across all days of 2024
• Policy start date = Purchase Date + 365 days
• Premium calculation rule:

Premium = Policy Tenure × 100

Purpose

This dataset represents the premium revenue generated by the insurance portfolio and serves as the base dataset for further claim analysis.

B. Claims Dataset

The Claims dataset represents insurance claims filed against policies based on predefined business rules.

Dataset Fields

• Claim_ID – Unique claim identifier
• Customer_ID – Customer associated with the claim
• Vehicle_ID – Vehicle associated with the claim
• Claim_Amount – Claim payout amount
• Claim_Date – Date the claim was filed
• Claim_Type – Indicates first or second claim

Claim Assumptions

• Claim Amount = 10% of Vehicle Value

Claim Rules – 2025

• Vehicles purchased on the 7th, 14th, 21st, and 28th of each month were eligible for claims.
• 30% of these vehicles were randomly selected to file a claim.
• Claim date was set as the policy start date to ensure the policy was active.

Claim Rules – 2026

• Between Jan 1 and Feb 28, 2026, additional claims were generated.
• 10% of vehicles with 4-year tenure filed claims.
• Claims were distributed evenly across the 59-day period.
• Vehicles that claimed in 2025 could file a second claim (Claim Type 2) in 2026 if eligible.

2. Data Preparation and Automation

After generating the datasets, an automated Python pipeline was implemented to clean and integrate the data for analysis.

Objective

Prepare a clean dataset for analytical queries and Power BI visualization.

Main Steps

• Load policy and claims datasets
• Merge datasets using Vehicle_ID
• Remove duplicate columns
• Handle missing values
• Convert date columns to datetime
• Create derived variables
• Perform validation checks
• Export the cleaned dataset

A. Dataset Integration

The datasets were merged to create a unified analytical dataset.

Method

• Join Type: **Left Join**
• Join Key: **Vehicle_ID**

This ensures all policies remain in the dataset, including vehicles without claims.

B. Handling Duplicate Columns

During the merge, duplicate identifiers appeared:

• Customer_ID_x
• Customer_ID_y

Cleaning steps:

• Removed Customer_ID_y
• Renamed Customer_ID_x → Customer_ID

C. Handling Missing Values

Missing values appeared for vehicles without claims.

Handling approach:

• Claim_ID → kept as NULL
• Claim_Date → kept as NULL
• Claim_Type → kept as NULL
• Claim_Amount → replaced with **0**

D. Feature Engineering

Additional variables were created for analysis.

Derived Columns

• **Claim_or_Not** – Binary indicator (1 = claim, 0 = no claim)
• **Purchase_Month** – Extracted from policy purchase date
• **Claim_Year & Claim_Month** – Derived from claim date
• **Tenure_Days** – Policy tenure converted to days

Tenure_Days = Policy_Tenure × 365

E. Data Validation

Validation checks ensured dataset accuracy.

• Verified dataset size matches policy records
• Checked for duplicate Vehicle_ID values
• Analyzed claim distribution

Results showed:

• Most vehicles filed no claims
• Some vehicles filed one claim
• Very few vehicles filed two claims

This confirms the simulation logic was applied correctly.

F. Final Dataset Export

The final cleaned dataset was exported as

cleaned_insurance_analytics.csv

This dataset is used for:

• Analytical queries
• Summary tables
• Power BI dashboard development

Analytical Queries and Business Insights

Part 1:

1. Calculate the total premium collected during the year 2024.

1. Total Premium Collected in 2024

```python
df["Policy_Purchase_Date"] = pd.to_datetime(df["Policy_Purchase_Date"])
```

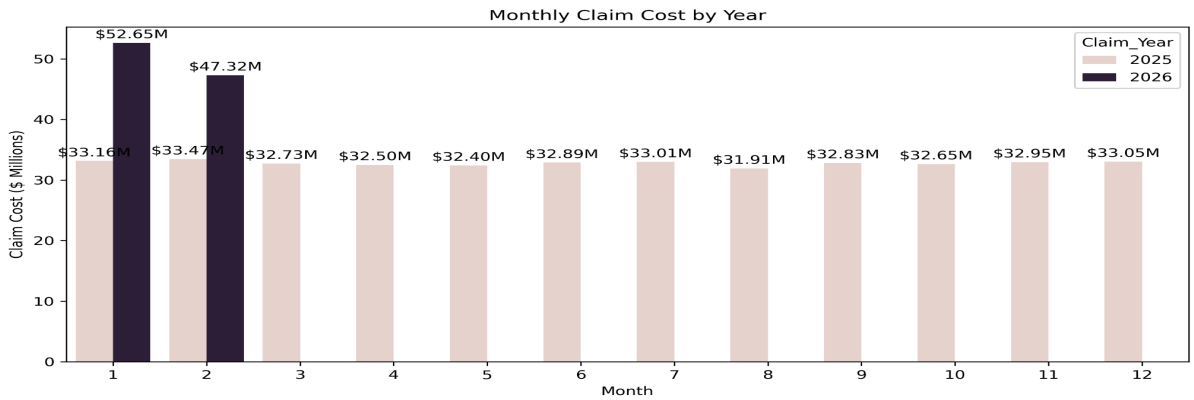
```python
premium_2024 = df[df["Policy_Purchase_Date"].dt.year == 2024]["Premium"].sum()
print(f"Total Premium Collected in 2024: ${premium_2024 :,.0f}")
```

```
Total Premium Collected in 2024: $240,254,000
```

2. Calculate the total claim cost for each year (2025 and 2026) with a monthly breakdown.

2. Calculate the total claim cost for each year (2025 and 2026) with a monthly breakdown.

```python
# Keep only rows where a claim exists
claims_df = df[df["Claim_Amount"].notna()]
# Monthly claim cost calculation
claims_summary = (
    claims_df
    .groupby(["Claim_Year", "Claim_Month"], as_index=False)["Claim_Amount"]
    .sum()
)
# Convert to integer
claims_summary["Claim_Year"] = claims_summary["Claim_Year"].astype(int)
claims_summary["Claim_Month"] = claims_summary["Claim_Month"].astype(int)
# Convert claim amount to millions
claims_summary["Claim_Amount_M"] = claims_summary["Claim_Amount"] / 1_000_000
# Format output in $M
claims_summary["Claim_Amount"] = claims_summary["Claim_Amount_M"].apply(
    lambda x: f"${x:,.2f}M")
# Final table
final_output = claims_summary[["Claim_Year", "Claim_Month", "Claim_Amount"]]
print(final_output)
```



Insights

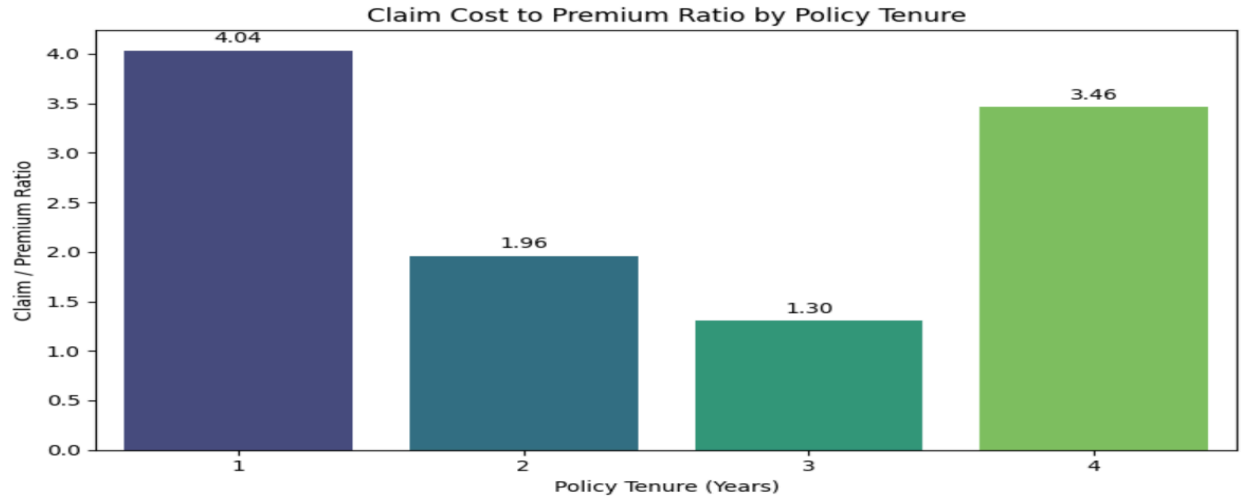
- Claim costs in 2025 remain consistent, with monthly values ranging between $31M and $33M.
- 2026 shows a sharp increase in the first two months, with claim costs exceeding $47M–$52M.
- This increase is driven by the additional claim simulation rule for long-tenure policies.
- The results indicate that future claim exposure can significantly impact yearly financial performance.

3. Calculate the claim cost to premium ratio for each policy tenure (1, 2, 3, and 4 years).

3. Calculate the claim cost to premium ratio for each policy tenure (1, 2, 3, and 4 years).

```python
# Calculate totals by policy tenure
tenure_analysis = (
    df.groupby("Policy_Tenure")
    .agg(
        Total_Premium=("Premium","sum"),
        Total_Claims=("Claim_Amount","sum")
    )
    .reset_index()
)
# Replace NaN claims with 0
tenure_analysis["Total_Claims"] = tenure_analysis["Total_Claims"].fillna(0)

# Calculate ratio
tenure_analysis["Claim_to_Premium_Ratio"] = (
    tenure_analysis["Total_Claims"] /
    tenure_analysis["Total_Premium"]
)
print(tenure_analysis)
```

Python



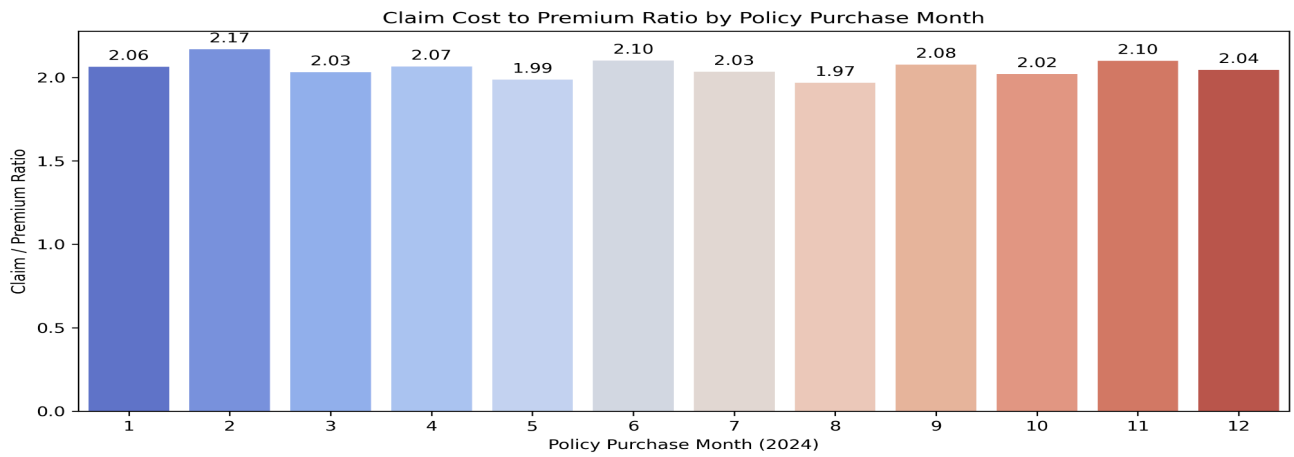
Insights

• 1-year policies show the highest loss ratio (~4.04), indicating claim costs significantly exceed the premium collected.
• 3-year policies have the lowest loss ratio (~1.30), making them the most profitable policy tenure in the portfolio.
• 4-year policies also show high loss exposure (~3.46) due to the longer coverage duration.
• The results suggest that medium-term policies (3 years) provide the best balance between revenue and risk.

4. Calculate the claim cost to premium ratio by the month in which the policy was sold (January–December 2024).

4. Calculate the claim cost to premium ratio by the month in which the policy was sold (January–December 2024).

```python
import pandas as pd

# Replace NaN claim values with 0
df["Claim_Amount"] = df["Claim_Amount"].fillna(0)

# Aggregate by purchase month
monthly_analysis = (
    df.groupby("Purchase_Month")
    .agg(
        Total_Premium=("Premium", "sum"),
        Total_Claims=("Claim_Amount", "sum")
    )
    .reset_index()
)

# Calculate ratio
monthly_analysis["Claim_to_Premium_Ratio"] = (
    monthly_analysis["Total_Claims"] /
    monthly_analysis["Total_Premium"]
)

print(monthly_analysis)
```



Insights

• The claim cost to premium ratio remains relatively stable across all purchase months, averaging around 2.0.

• February shows the highest ratio (~2.17), indicating slightly higher claim cost relative to premium for policies purchased in that month.

• August shows the lowest ratio (~1.97), suggesting marginally lower claim risk.

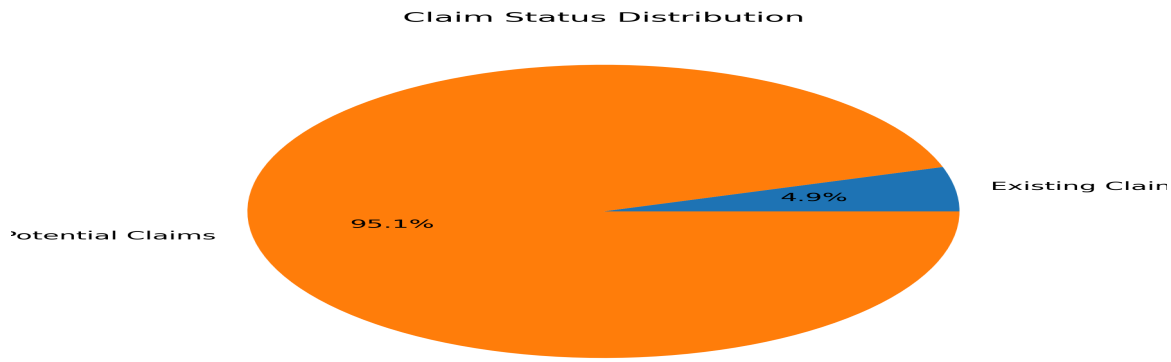
• Overall, the variation across months is minimal, indicating that policy purchase month does not significantly impact portfolio profitability or claim risk.

5. If every vehicle that has not yet made a claim eventually files exactly one claim during the remaining policy tenure, estimate the total potential claim liability.

```python
# Claim amount per vehicle
claim_value = 10000

# Vehicles that never made a claim
no_claim_df = df[df["Claim_or_Not"] == 0]

# Count vehicles
no_claim_count = no_claim_df["Vehicle_ID"].nunique()

# Potential liability
total_liability = no_claim_count * claim_value

print(f"Vehicles with no claims yet: {no_claim_count:,}")
print(f"Estimated Future Claim Liability: ${total_liability:,.0f}")
liability_million = total_liability / 1_000_000

print(f"Estimated Future Claim Liability: ${liability_million:,.2f}M")
```

Python

```
Vehicles with no claims yet: 951,006
Estimated Future Claim Liability: $9,510,060,000
Estimated Future Claim Liability: $9,510.06M
```



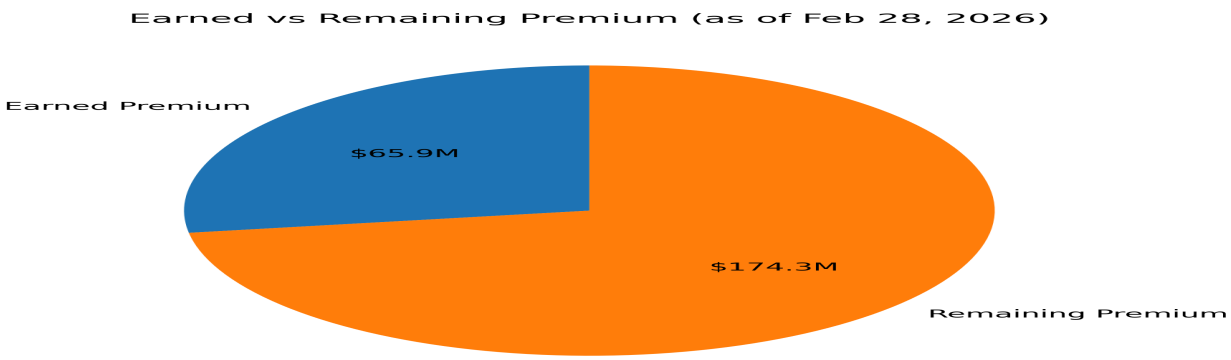
Insights

• Only 4.9% of vehicles have filed claims, while 95.1% remain potential claims, indicating significant future claim liability for the insurer.

• This suggests the portfolio still carries high future risk exposure during the remaining policy tenure.

6. Assume daily premium = Total Premium ÷ Total Policy Tenure Days. Based on this: • Calculate the premium already earned by the company up to February 28, 2026. • Estimate the premium expected to be earned monthly for the remaining policy period (assume 46 months remaining).

```python
# Cutoff Date
cutoff_date = pd.to_datetime("2026-02-28")
# Daily Premium
df["Daily_Premium"] = df["Premium"] / df["Tenure_Days"]
# Earned Days
df["Earned_Days"] = (cutoff_date - df["Policy_Start_Date"]).dt.days
df["Earned_Days"] = df["Earned_Days"].clip(lower=0)
df["Earned_Days"] = df[["Earned_Days","Tenure_Days"]].min(axis=1)
# Earned Premium
df["Earned_Premium"] = df["Daily_Premium"] * df["Earned_Days"]
earned_premium_total = df["Earned_Premium"].sum()
print(f"Premium Earned until Feb 28 2026: ${earned_premium_total:,.2f}")
# Remaining Premium
total_premium = df["Premium"].sum()
remaining_premium = total_premium - earned_premium_total
# Monthly Expected Premium
monthly_future_premium = remaining_premium / 46
print(f"Remaining Premium: ${remaining_premium:,.2f}")
print(f"Expected Monthly Premium (Next 46 Months): ${monthly_future_premium:,.2f}")
# Convert to Millions
earned_m = earned_premium_total / 1_000_000
monthly_m = monthly_future_premium / 1_000_000
print(f"Earned Premium: ${earned_m:.2f}M")
print(f"Expected Monthly Premium: ${monthly_m:.2f}M")
```



Insights

• As of Feb 28, 2026, only $65.9M premium has been earned, while $174.3M remains unearned, meaning most revenue will be recognized in the future.

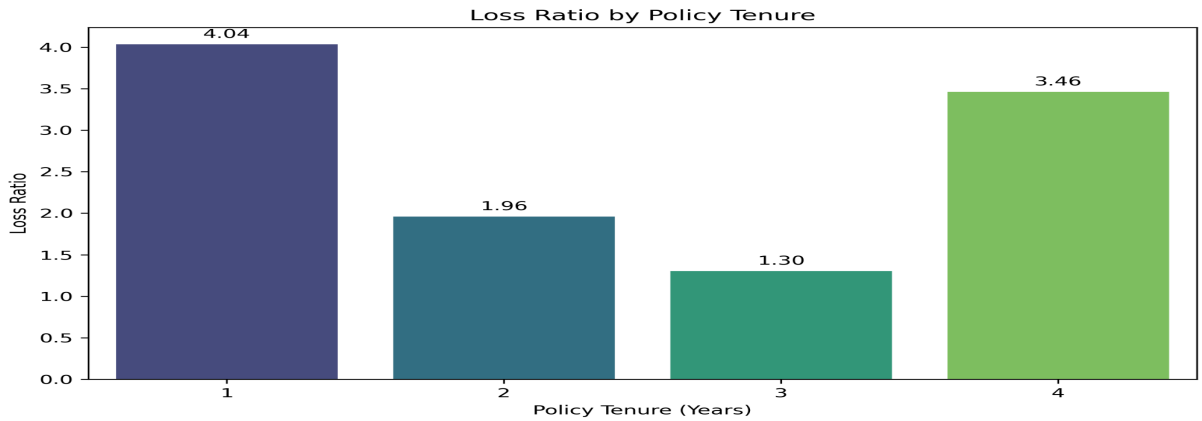
• This reflects the long-term nature of insurance policies where premium revenue is earned gradually over time.

Part 2

1. Identify which policy tenure appears most profitable and explain why.

Q1. Identify which policy tenure appears most profitable and explain why.

```python
# Replace NaN claims with 0
df["Claim_Amount"] = df["Claim_Amount"].fillna(0)

# Calculate totals by tenure
tenure_profit = (
    df.groupby("Policy_Tenure")
    .agg(
        Total_Premium=("Premium","sum"),
        Total_Claims=("Claim_Amount","sum")
    )
    .reset_index()
)

# Loss ratio
tenure_profit["Loss_Ratio"] = (
    tenure_profit["Total_Claims"] /
    tenure_profit["Total_Premium"]
)

print(tenure_profit)
```



Insights

•The 3-year policy tenure is the most profitable because it balances premium revenue and claim exposure effectively.

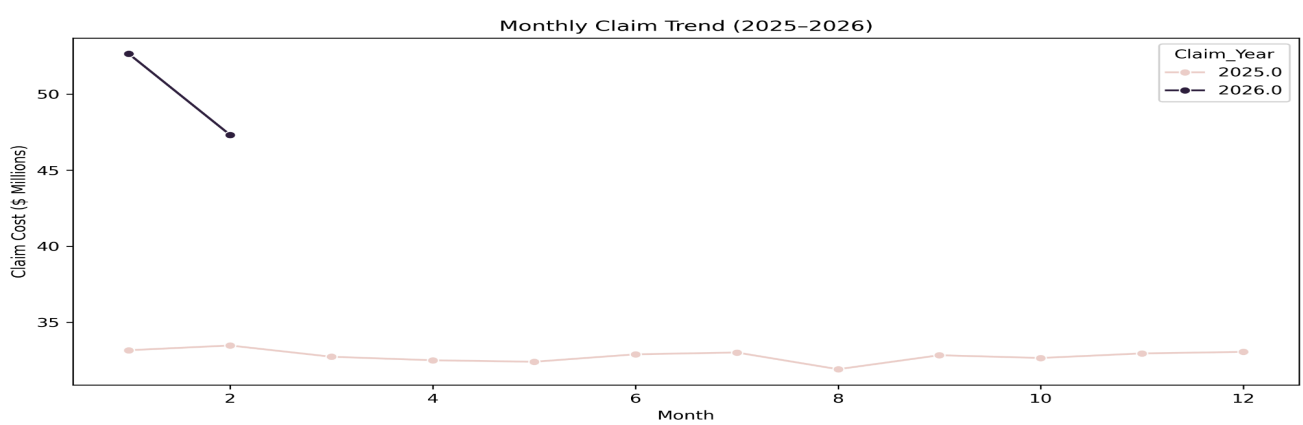
•It has the lowest claim-to-premium ratio (1.30), while 1-year policies generate lower premiums and 4-year policies face higher claim risk due to longer coverage periods.

2. Build a simple dashboard or visualization showing claim trends by month.

Q2. Build a simple dashboard or visualization showing claim trends by month.

```python
import matplotlib.pyplot as plt
import seaborn as sns
import os
os.makedirs("images", exist_ok=True)
plt.figure(figsize=(10,5))
sns.lineplot(
    data=claims_summary,
    x="Claim_Month",
    y="Claim_Amount_M",
    hue="Claim_Year",
    marker="o"
)
plt.xlabel("Month")
plt.ylabel("Claim Cost ($ Millions)")
plt.title("Monthly Claim Trend (2025–2026)")
plt.tight_layout()
plt.savefig("images/claim_trend_by_month.png", dpi=300)
plt.show()
```

[40]



Insights

•Claim costs remain relatively stable throughout 2025, indicating a consistent pattern of claims due to evenly distributed policy sales.
•In 2026, claims appear only in January and February, with higher costs compared to individual months in 2025 because 4-year tenure policies generate additional claims during this period.

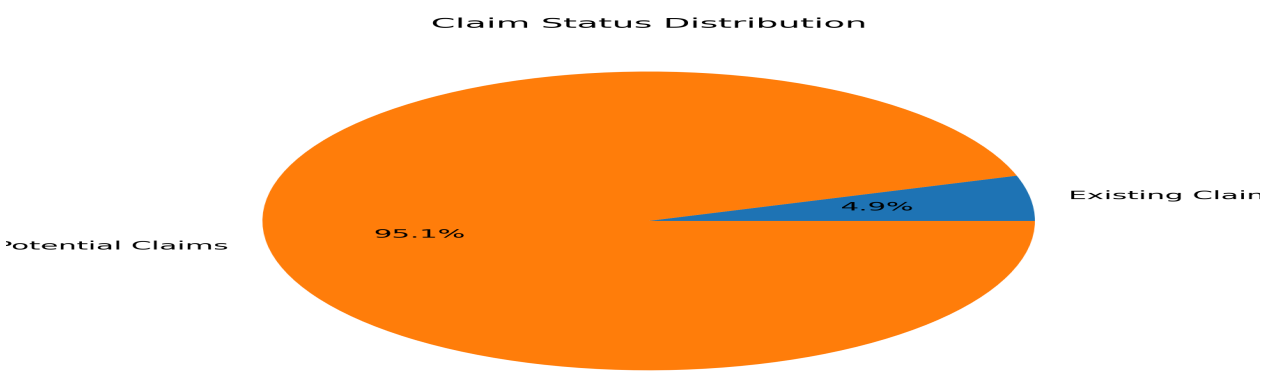
3. Estimate the loss ratio (Claims ÷ Premium) for the portfolio.

Q3. Estimate the loss ratio (Claims ÷ Premium) for the portfolio.

```python
# Replace NaN claims with 0
df["Claim_Amount"] = df["Claim_Amount"].fillna(0)

# Calculate totals
total_claims = df["Claim_Amount"].sum()
total_premium = df["Premium"].sum()

# Loss ratio
portfolio_loss_ratio = total_claims / total_premium

premium_m = total_premium / 1_000_000
claims_m = total_claims / 1_000_000

print(f"Total Premium: ${premium_m:.2f}M")
print(f"Total Claims: ${claims_m:.2f}M")
print(f"Portfolio Loss Ratio: {portfolio_loss_ratio:.2f}")
```

```
Total Premium: $240.25M
Total Claims: $493.52M
Portfolio Loss Ratio: 2.05
```

Insights
• The portfolio shows a loss ratio of 2.05, meaning total claims are more than double the premium collected.
• This indicates that the portfolio is currently unprofitable, as claim payouts significantly exceed premium revenue.



Insights
• Only 4.9% of vehicles have filed claims, while 95.1% remain potential claims, indicating very low observed claim frequency so far.

• This suggests the insurer still faces significant future claim liability during the remaining policy tenure.

4. If claim frequency increases by 5% annually, estimate the impact on future profitability.

4. If claim frequency increases by 5% annually, estimate the impact on future profitability.

```python
# Replace NaN claims with 0
df["Claim_Amount"] = df["Claim_Amount"].fillna(0)
# Current totals
total_premium = df["Premium"].sum()
total_claims = df["Claim_Amount"].sum()
# Current loss ratio
current_loss_ratio = total_claims / total_premium
print(f"Current Loss Ratio: {current_loss_ratio:.2f}")
```

```
Current Loss Ratio: 2.05
```

```python
# Increase claims by 5%
future_claims = total_claims * 1.05
# New loss ratio
future_loss_ratio = future_claims / total_premium
print(f"Future Loss Ratio after 5% increase in claims: {future_loss_ratio:.2f}")
```

```
Future Loss Ratio after 5% increase in claims: 2.16
```

```python
loss_ratio_change = future_loss_ratio - current_loss_ratio

print(f"Current Loss Ratio: {current_loss_ratio:.2f}")
print(f"Future Loss Ratio (after 5% claim increase): {future_loss_ratio:.2f}")
print(f"Increase in Loss Ratio: {loss_ratio_change:.2f}")
```

[46]

```
Current Loss Ratio: 2.05
Future Loss Ratio (after 5% claim increase): 2.16
Increase in Loss Ratio: 0.10
```

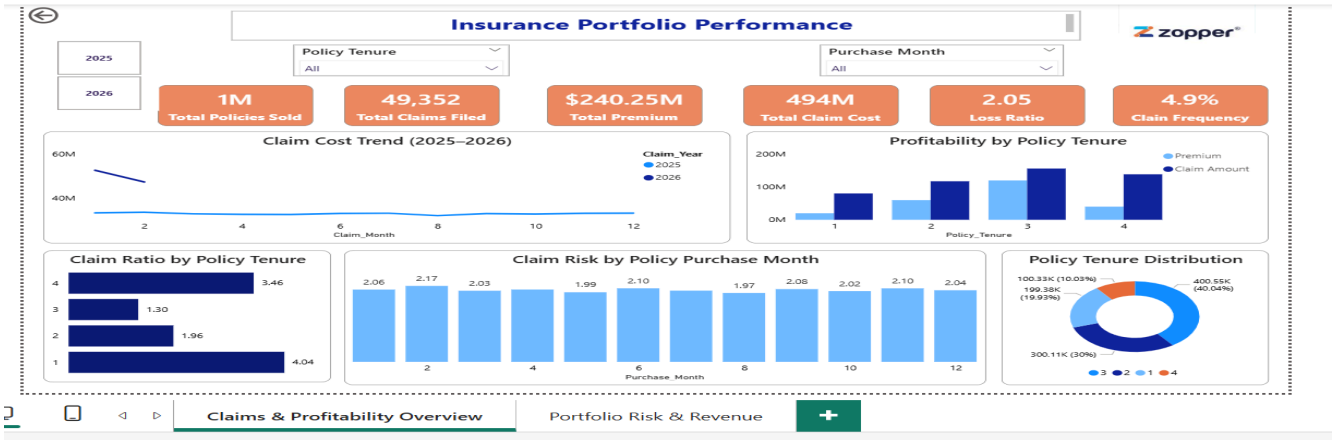
Insights

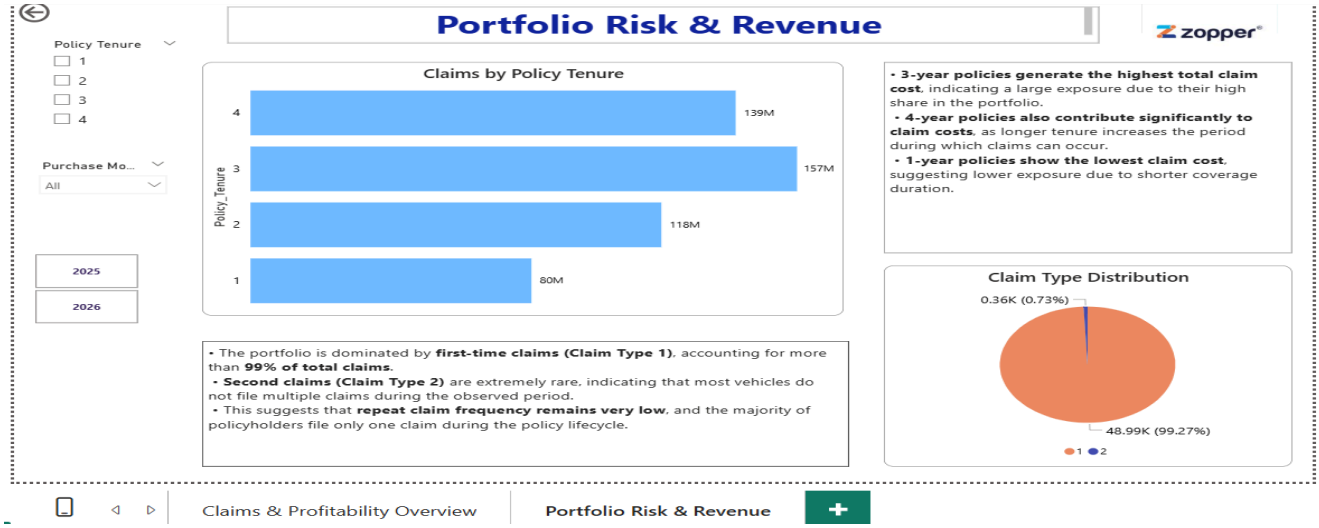
A 5% increase in claim frequency raises the portfolio loss ratio from 2.05 to 2.16, indicating that claims consume an even larger portion of premium revenue. This suggests a decline in overall profitability.

Recommendation:

The insurer should consider adjusting premium pricing, strengthening risk assessment, or improving claim management practices to mitigate the financial impact of increasing claim frequency.

Insurance Analysis Dashboard





Key Insights From Dashboard Analysis:

1. The portfolio consists of approximately 1 million policies with a total premium collection of approximately $240 million.

2. The analysis shows that claim frequency remains relatively low at around 4.9%, indicating that only a small portion of policies result in claims.

3. Total claim cost across the observed period amounts to approximately $494 million, resulting in a portfolio loss ratio of approximately 2.05.

4. Policy tenure plays a significant role in claim exposure. Policies with longer tenures contribute more to total claim costs due to the extended coverage period.

5. The majority of claims are first-time claims (Claim Type 1), accounting for more than 99% of total claims. Second claims are extremely rare, indicating that most vehicles file only one claim during the observed period.

6. Monthly claim trends show relatively stable claim costs throughout the year, suggesting that claim occurrences are distributed consistently over time rather than concentrated in specific months.

BUSINESS RECOMMENDATIONS

Based on the analysis of policy sales and claim behavior, the following recommendations are proposed to improve portfolio profitability and manage insurance risk more effectively.

1. Optimize Pricing Strategy for High-Risk Tenures

The analysis shows that longer policy tenures generate higher claim exposure. In particular, 3-year and 4-year policies contribute significantly to total claim costs. Insurance providers should consider adjusting premium pricing for longer tenure policies to better reflect the increased risk and duration of coverage.

2. Strengthen Risk Monitoring for Long-Term Policies

Policies with longer tenures remain active for extended periods, increasing the probability of claims. Continuous monitoring of these policies and periodic risk assessment can help insurers manage claim exposure more effectively.

3. Improve Predictive Risk Modeling

Using historical policy and claim data, insurers can implement predictive analytics models to estimate the likelihood of claims. Machine learning models could be used to identify high-risk customers and adjust underwriting strategies accordingly.

4. Monitor Claim Trends by Policy Purchase Month

The dashboard analysis highlights variations in claim ratios across different purchase months. Tracking seasonal patterns in claim frequency can help insurers better anticipate risk and allocate resources accordingly.

5. Encourage Shorter Policy Tenures for Lower Risk Segments

Shorter tenure policies show lower overall claim exposure. Offering incentives or discounts for shorter policy durations could help maintain a balanced risk portfolio.

6. Develop Early Warning Indicators for Claim Risk

By monitoring policy characteristics such as tenure, purchase period, and claim history, insurers can develop early warning indicators that help identify potentially high-risk policies before claims occur.